TRBD Consulting Group: Tech Stack

Status: final, agreed 21 Oct 2026. Prices are approximate list prices as of mid-2026 and should be confirmed on each vendor's pricing page before signing up.

Overview

TRBD Consulting Group Private Limited is an India-registered consultancy working across five verticals: Manpower, Education & Training, Oil & Gas, Green Energy and Global Business. The website starts as a picture-rich, SEO-focused landing page and grows into a platform for vertical-routed enquiries, user registration, Manpower candidate profiles with CVs, Training courses and enrolments, and an insights section.

The system is a **single Next.js application** that serves both the public pages and the backend logic. It is hosted on **Vercel**, with **Supabase** providing the Postgres database, user sign-in and file storage. **Payload CMS** runs inside the same app, so the content writer can edit text and images from a web editor without touching the codebase, and staff use the same admin to work enquiries. All data is held in the **Mumbai** region, and India's **Digital Personal Data Protection (DPDP) Act 2023** is the baseline for consent, purpose limits and deletion.

Guiding principles: one codebase and one language (TypeScript), low fixed cost, very little server upkeep, portable standard technology (Postgres, open-source CMS), and good documentation.

Summary

Area	Choice	Why	Cost per month
Framework	Next.js 16 (App Router), React 19, TypeScript	Already in place; full-stack in one codebase; strong SEO features	Free
Styling	Tailwind CSS 4, `lucide-react` icons	Already in place; fast to build the redesign	Free
Hosting	Vercel Pro, functions in Mumbai (`bom1`)	Native Next.js support, global CDN, image optimisation, a preview link for every change; commercial use allowed	$20 (1 seat)
Database	Supabase Postgres, Mumbai (`ap-south-1`)	Managed Postgres with daily backups and row-level security	Free, then $25 (Pro)
ORM and migrations	Drizzle ORM	Typed queries and versioned SQL migrations kept in the repo	Free

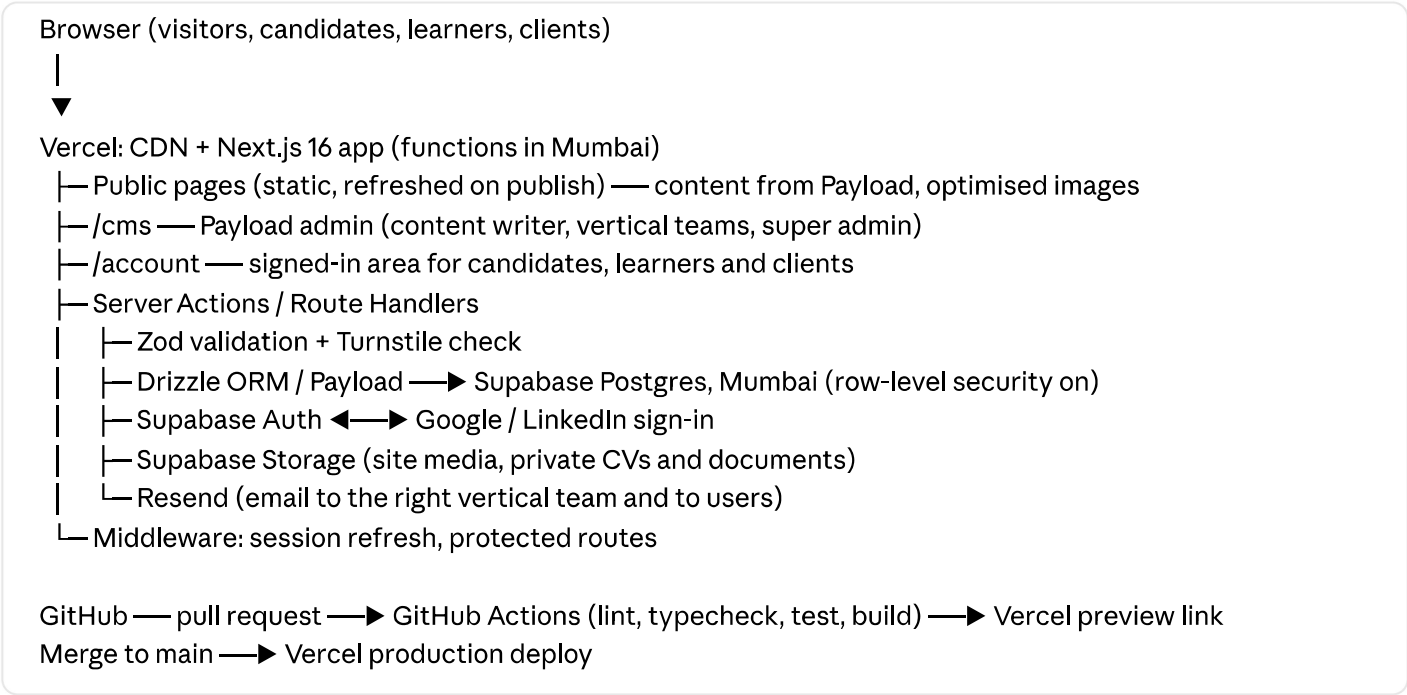
Area	Choice	Why	Cost per month
User sign-in	Supabase Auth: email magic link, Google, LinkedIn	For candidates, learners and clients; no auth code to maintain	Included
CMS and back office	Payload CMS 3, inside the app at `/cms`	Content writer edits pages, images and articles with live preview; staff manage enquiries per vertical	Free
File storage	Supabase Storage: public bucket for site media, private bucket for CVs and documents	One place for all files, with access rules and expiring links for sensitive documents	Included
Validation	Zod	One schema shared by forms, server code and the database layer	Free
Email	Resend with React Email templates	Enquiry notifications routed by vertical, welcome and verification emails	Free to 3,000/mo, then $20

Area	Choice	Why	Cost per month
Spam protection	Cloudflare Turnstile	Free, privacy-friendly CAPTCHA on all public forms	Free
Images	`next/image` with Vercel's image CDN	Fast loading for the picture-heavy design	Included
Analytics	Vercel Web Analytics	Cookie-free, so no consent banner needed for analytics	Included
Error monitoring	Sentry	Alerts on server and browser errors	Free tier
Testing	Vitest (unit), Playwright (end-to-end)	Standard Next.js testing tools	Free
CI	GitHub Actions: lint, typecheck, tests, build on every pull request	Catches problems before they reach the live site	Free
SEO	Next.js Metadata API, `sitemap.ts`, `robots.ts`, JSON-LD, Open Graph images, one page per vertical	Built into the framework; SEO fields editable in the CMS	Free
Documentation	`README.md`, `docs/architecture.md`, decision records in `docs/adr/`	Keeps the project easy to hand over	Free

Area	Choice	Why	Cost per month
Compliance	DPDP Act 2023; data kept in Mumbai	India-registered company handling personal data	n/a

Expected running cost: about $20 a month while the site is a landing page, and about $45 to $65 a month once registration and CV uploads go live (Supabase Pro, and Resend if email volume grows). A domain costs around $15 a year, and business email such as Google Workspace is separate.

Architecture



Content publishing: when the content writer publishes in Payload, the affected pages are refreshed automatically within seconds. No developer or redeploy is needed.

Who signs in where: staff (content writer, vertical teams, super admin) sign in to Payload at `/cms`. Public users (candidates, learners, clients) sign in with Supabase Auth at `/account`.

CMS content model

The page layout is built from fixed section blocks that the content writer can fill in, reorder or hide, but cannot break, so the design stays intact.

- **Site settings (global):** logo, contact email, phone, office address, working hours, social links, default SEO image
- **Navigation (global):** header and footer links
- **Pages:** title, slug, SEO fields (meta title, description, Open Graph image), and section blocks: Hero, Verticals grid, Approach stages, Why TRBD points, Image and text, Stats, Testimonials, Call to action, Contact

- **Verticals:** name, slug, summary, full description, hero image, services list, SEO fields, team inbox email used for enquiry routing
- **Insights:** title, slug, author, vertical tags, cover image, rich text body, publish date, SEO fields
- **Media:** image, alt text (required for accessibility and SEO), caption, focal point for cropping

Editors get drafts, version history with rollback, scheduled publishing and live preview.

Data model

Leads and accounts:

- `verticals` : id, slug (`manpower` , `education-training` , `oil-gas` , `green-energy` , `global-business`), name, notification_email
- `enquiries` : id, full_name, company, email, phone, requirement, source_page, status (`new` , `contacted` , `qualified` , `won` , `lost`), assigned_to, created_at, first_response_at (tracks the two-working-day reply promise)
- `enquiry_verticals` : enquiry_id, vertical_id (an enquiry for "More than one" vertical has several rows)
- `profiles` : id (same as the sign-in user id), name, company, phone, country, created_at
- `consents` : user or enquiry, type (privacy notice version, marketing), granted_at, withdrawn_at

Vertical modules (added as needed):

- Manpower: `candidates` (profile, skills, trade, years of experience, certifications, passport and visa status, preferred countries), `candidate_documents` (file path, type, uploaded_at), `job_openings` (client, vertical, location, positions), `applications`
- Education & Training: `courses` , `course_batches` (dates, mode, seats), `enrolments`

Roles and access

Role	Signs in at	Can do
Super admin	`/cms`	Everything, including users, roles and settings
Content editor	`/cms`	Edit and publish pages, verticals, insights and media
Vertical team	`/cms`	See and work only their own vertical's enquiries, candidates or courses
Candidate / learner / client	`/account`	Manage their own profile, documents, applications and enrolments

Access rules are enforced in the database (row-level security) as well as in the app. CVs and identity documents sit in a private bucket and are shared only through short-lived links.

Security and data protection

- All personal data stored in Mumbai; backups in the same region.

- Consent captured with the privacy notice version at every form and sign-up; users can withdraw consent, export their data and request deletion.
- Collect only what each form needs; set retention periods for enquiries and candidate documents.
- Secrets kept in Vercel and GitHub environment settings, never in the code.
- Turnstile and rate limiting on every public form.
- If many candidates or clients are in the EU, UK or Gulf, check whether GDPR or local rules also apply to their data.

Roadmap

1. **Foundations (alongside the redesign):** set up Vercel Pro and Supabase in Mumbai, install Payload with the content model above and move the landing page content into it, add CI, linting, typechecks and the test setup, SEO metadata, sitemap, robots, JSON-LD and one page per vertical.
2. **Enquiries:** replace Web3Forms with our own form that checks input, blocks spam, saves each enquiry with its verticals and emails the right team. Teams work their enquiries in Payload.
3. **Registration:** Supabase Auth sign-in, profiles, the `/account` area, consent capture and a privacy policy page.
4. **Manpower module:** candidate registration with CV and document upload, job openings and applications.
5. **Training module:** courses, batches and enrolments, with payments later if needed.
6. **Insights:** articles section published from Payload.

Notes

- Confirm that the Payload version being installed supports Next.js 16.
- Decide who besides the content writer needs CMS access, so roles can be set up from the start.